

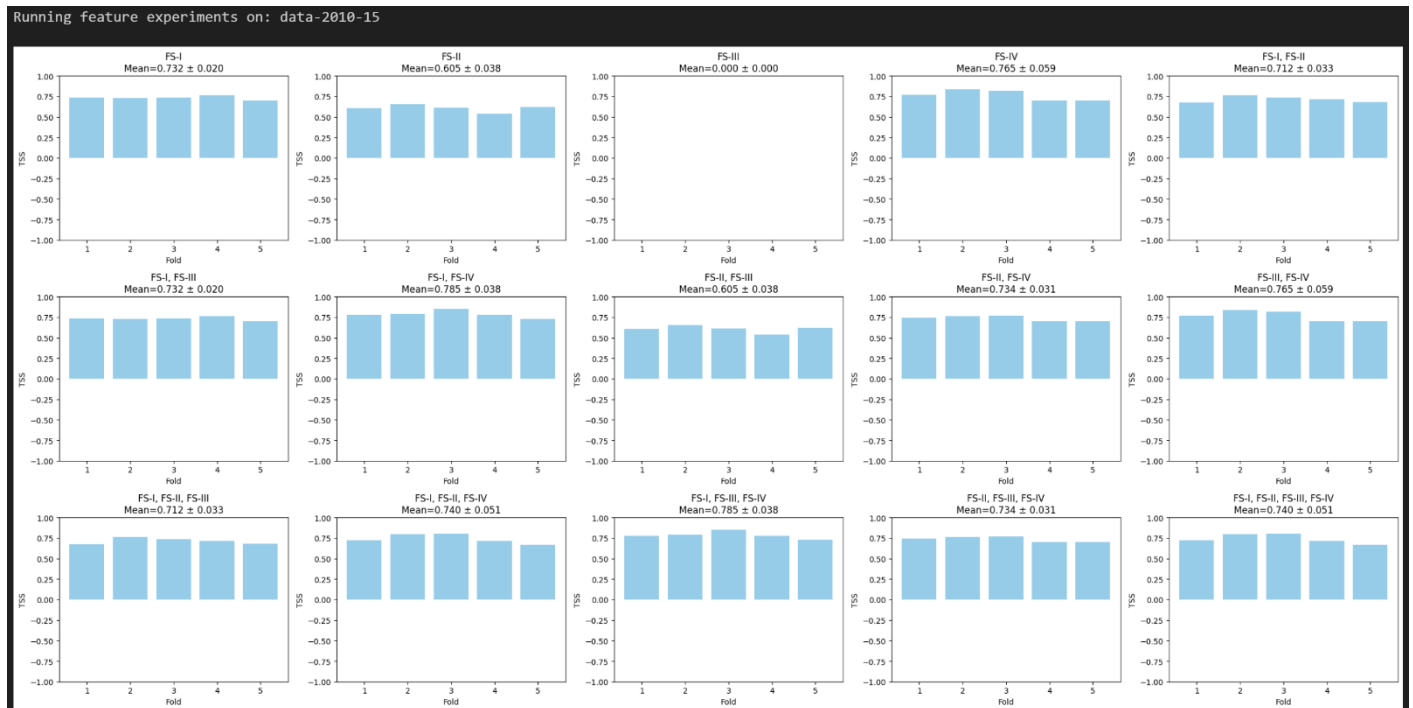
Report Overview

This report summarizes the findings from Assignment 2 of the CompSci 4AL3 course. This project implements a Support Vector Machine (SVM) model to predict the occurrence of solar flares using multiple feature sets from NASA's solar data. The program loads positive and negative event samples, preprocesses and normalizes them, and evaluates the model through k-fold cross-validation ($k = 5$). Performance is measured using the True Skill Statistic (TSS), with results visualized through bar charts and confusion matrices. The experiments are run on two time-based datasets (2010–2015 and 2020–2024) to compare model accuracy and feature effectiveness across different solar cycles. NOTE: FS-III did not have any data associated with it which is why it is blank in the data.

Dataset 2010-2015

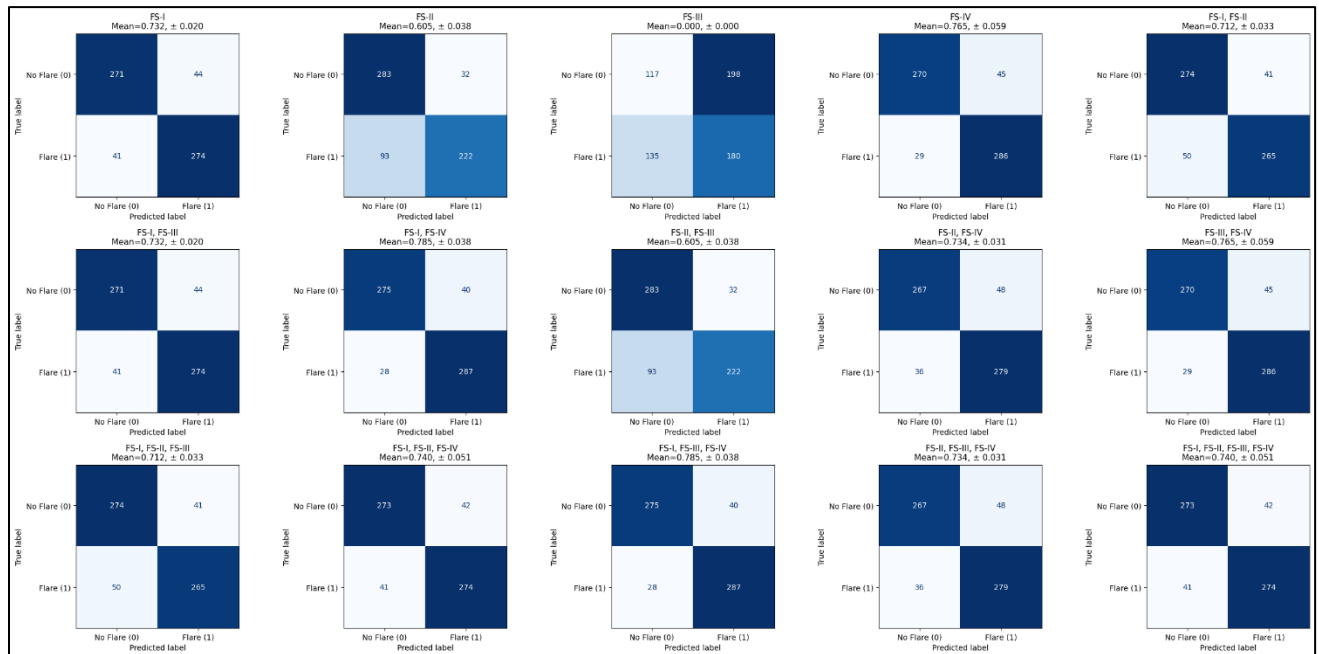
1. Bar chart showing performance of each model set feature combination across each fold

Below is a figure displaying various bar charts for each feature combination in the 2010-2015 dataset. The TSS means and associated standard deviations are also displayed for each fold across each combo.



2. Confusion matrices of each model set feature combination

Below is a figure displaying the confusion matrices for each feature combination in the 2010-2015 dataset. The matrices are organized in terms of TP (true positive), FP (false negative), FN (false negative), and TN (true negative). We can see that the best performing feature combination was FS-I, FS-IV (also FS-I, FS-III, FS-IV are equivalent since FS-III has no data). This was achieved using the RBF kernel, $C = 1$ (for less overfitting) and gamma = 'scale' (to adapt to data accordingly). After multiple trials with different kernels and C values, this proved to produce one of the highest mean TSS scores on average across all the combinations.



3. All combinations means and standard deviations

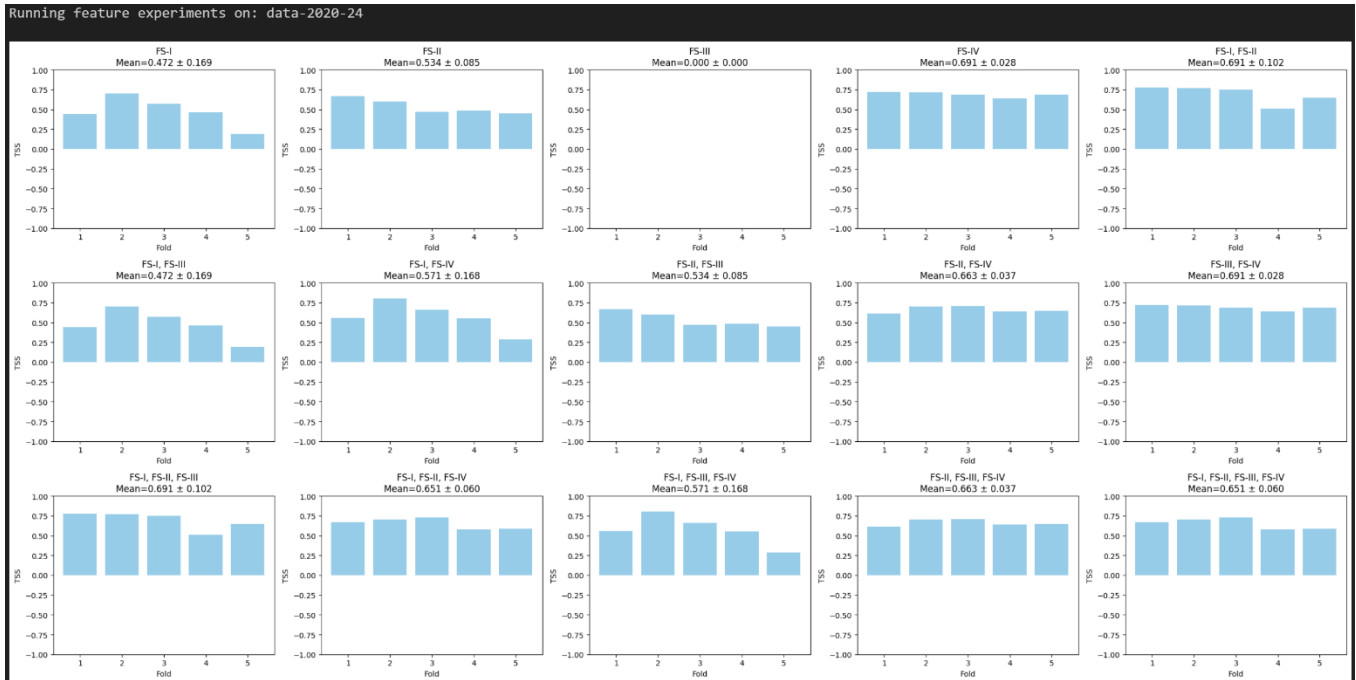
Below is the output for all 15 combinations' means and standard deviations.

```
Feature combo: FS-I | Mean TSS: 0.7317 | Std TSS: 0.0199
Feature combo: FS-II | Mean TSS: 0.6045 | Std TSS: 0.0382
Feature combo: FS-III | Mean TSS: 0.0000 | Std TSS: 0.0000
Feature combo: FS-IV | Mean TSS: 0.7652 | Std TSS: 0.0587
Feature combo: FS-I, FS-II | Mean TSS: 0.7124 | Std TSS: 0.0332
Feature combo: FS-I, FS-III | Mean TSS: 0.7317 | Std TSS: 0.0199
Feature combo: FS-I, FS-IV | Mean TSS: 0.7851 | Std TSS: 0.0384
Feature combo: FS-II, FS-III | Mean TSS: 0.6045 | Std TSS: 0.0382
Feature combo: FS-II, FS-IV | Mean TSS: 0.7341 | Std TSS: 0.0307
Feature combo: FS-III, FS-IV | Mean TSS: 0.7652 | Std TSS: 0.0587
Feature combo: FS-I, FS-II, FS-III | Mean TSS: 0.7124 | Std TSS: 0.0332
Feature combo: FS-I, FS-II, FS-IV | Mean TSS: 0.7399 | Std TSS: 0.0515
Feature combo: FS-I, FS-III, FS-IV | Mean TSS: 0.7851 | Std TSS: 0.0384
Feature combo: FS-II, FS-III, FS-IV | Mean TSS: 0.7341 | Std TSS: 0.0307
Feature combo: FS-I, FS-II, FS-III, FS-IV | Mean TSS: 0.7399 | Std TSS: 0.0515
```

Dataset 2020-2024

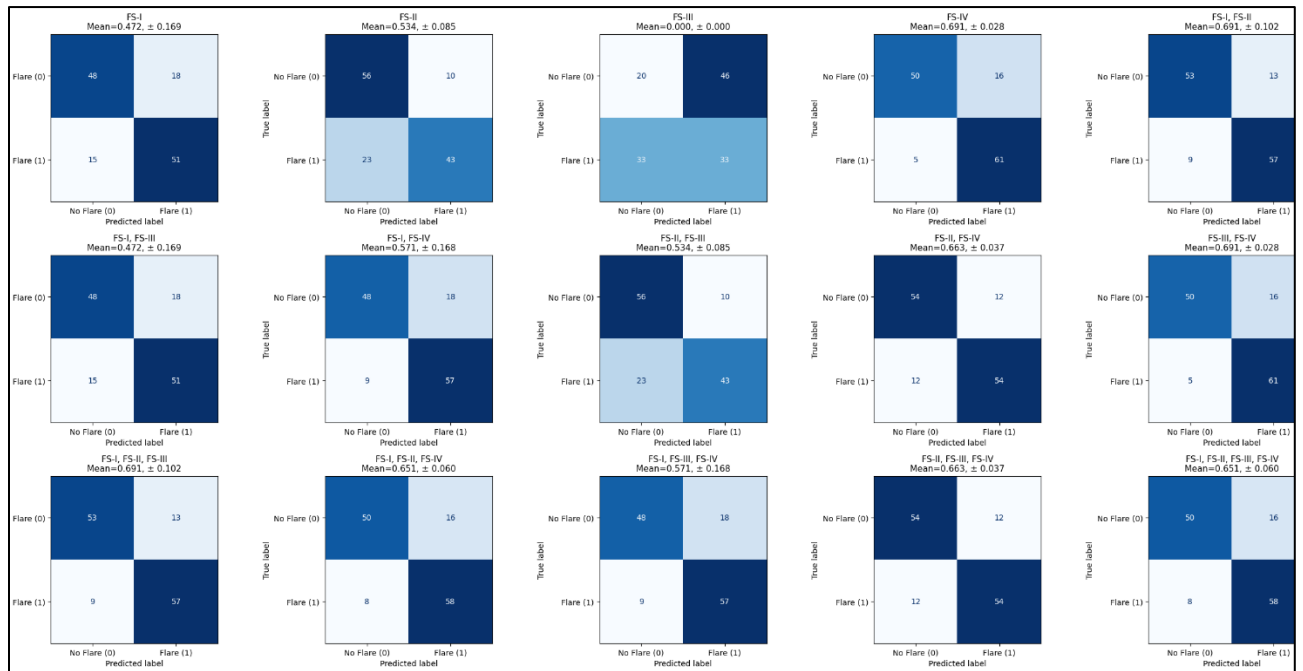
1. Bar chart showing performance of each model set feature combination across each fold

Below is a figure displaying various bar charts for each feature combination in the 2020-2024 dataset. The TSS means and associated standard deviations are also displayed for each fold across each combo.



2. Confusion matrices of each model set feature combination across both datasets

Below is a figure displaying the confusion matrices for each feature combination in the 2020-2024 dataset. The matrices are organized in terms of TP (true positive), FP (false negative), FN (false negative), and TN (true negative). We can see that the best performing feature combination was FS-I, FS-II (also FS-I, FS-II, FS-III are equivalent since FS-III has no data). This was achieved using the RBF kernel, $C = 1$ (for less overfitting) and $\gamma = \text{'scale'}$ (to adapt to data accordingly). After multiple trials with different kernels and C values, this proved to produce one of the highest mean TSS scores on average across all the combinations.



3. All combinations means and standard deviations

Below is the output for all 15 combinations' means and standard deviations.

```
Feature combo: FS-I | Mean TSS: 0.4724 | Std TSS: 0.1693
Feature combo: FS-II | Mean TSS: 0.5336 | Std TSS: 0.0848
Feature combo: FS-III | Mean TSS: 0.0000 | Std TSS: 0.0000
Feature combo: FS-IV | Mean TSS: 0.6907 | Std TSS: 0.0284
Feature combo: FS-I, FS-II | Mean TSS: 0.6910 | Std TSS: 0.1017
Feature combo: FS-I, FS-III | Mean TSS: 0.4724 | Std TSS: 0.1693
Feature combo: FS-I, FS-IV | Mean TSS: 0.5710 | Std TSS: 0.1682
Feature combo: FS-II, FS-III | Mean TSS: 0.5336 | Std TSS: 0.0848
Feature combo: FS-II, FS-IV | Mean TSS: 0.6625 | Std TSS: 0.0368
Feature combo: FS-III, FS-IV | Mean TSS: 0.6907 | Std TSS: 0.0284
Feature combo: FS-I, FS-II, FS-III | Mean TSS: 0.6910 | Std TSS: 0.1017
Feature combo: FS-I, FS-II, FS-IV | Mean TSS: 0.6514 | Std TSS: 0.0603
Feature combo: FS-I, FS-III, FS-IV | Mean TSS: 0.5710 | Std TSS: 0.1682
Feature combo: FS-II, FS-III, FS-IV | Mean TSS: 0.6625 | Std TSS: 0.0368
Feature combo: FS-I, FS-II, FS-III, FS-IV | Mean TSS: 0.6514 | Std TSS: 0.0603
```

Report Questions

1. Which feature combination worked best and which feature set was worst? Report the performance.

For the 2010-2015 dataset, the best feature combination set was FS-I, FS-IV (also FS-I, FS-III, FS-IV are equivalent since FS-III has no data) with Avg TSS = 0.7851 ± 0.0384 . The worst feature combination set was FS-II (also FS-II, FS-III since FS-III has no data) with Avg TSS = 0.605 ± 0.038 .

For the 2020-2024 dataset, the best feature combination set was FS-I, FS-II (also FS-I, FS-II, FS-III are equivalent since FS-III has no data) with Avg TSS = 0.6910 ± 0.1017 . The worst feature combination set was FS-I (also FS-I, FS-III since FS-III has no data) with Avg TSS = 0.472 ± 0.169 .

2. Does adding additional feature sets improve the TSS score? What do you observe with each feature set?

Adding additional feature sets did not consistently improve the TSS score. In the 2010–2015 dataset, individual feature sets such as FS-I and FS-IV already achieved relatively high TSS values (around 0.73–0.76), and while combining them sometimes provided a small increase (FS-I, FS-IV reached about 0.79), other combinations actually reduced performance. A similar trend was observed for the 2020–2024 dataset, where FS-IV again produced the highest individual performance (around 0.69), but adding more feature sets yielded mixed or even lower results. This suggests that FS-IV contains most of the predictive information, while the inclusion of additional features likely introduces redundancy or noise rather than improving model skill. In general, more features do not necessarily lead to better results unless they contribute some new unique or complementary information.

3. Which dataset led to a better TSS score (2010 or 2020)? Why do you think this is the case?

The SVM model achieved higher overall TSS values on the 2010–2015 dataset compared to the 2020–2024 dataset, with the best combination reaching about 0.78 versus roughly 0.69, respectively. This indicates that the model performed better on the earlier dataset. The difference is likely due to variations in data quality and distribution between solar cycles. The 2010–2015 period likely contained more frequent and well characterized flaring events, providing more balanced dataset for training. In contrast, the 2020–2024 dataset may have had fewer strong flares and potentially less consistent labeling, making it more difficult for the model to generalize. Additionally, the 2010-2015 dataset had a lot more values as compared to the 2020-2024 dataset which would allow the SVM to train more accurately on the prior dataset. As a result, the 2010–2015 dataset produced more stable and accurate TSS outcomes.